

Math 53 Cheat Sheet

10.1) Curves Defined by Parametric Eqns

parametric eq of a cycloid: $x = r(\theta - \sin \theta)$ $y = r(1 - \cos \theta)$

10.2) Calculus with Parametric Curves

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \quad \frac{d^2y}{dx^2} = \frac{d(\frac{dy}{dx})}{\frac{dx}{dt}} \quad A = \int_a^b y dx = \int_a^b g(t) f'(t) dt \quad \begin{cases} x = f(t), a = f(\alpha) \\ y = g(t), b = f(\beta) \end{cases} \quad L = \int_a^b \sqrt{1 + (\frac{dy}{dx})^2} dx = \int_a^b \sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2} dt$$

$$y - y_0 = f'(x_0)(x - x_0) \quad SA = \int_a^b 2\pi y ds = \int_a^b 2\pi y \sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2} dt \quad \text{y: rotation around x-axis (use x for y-axis)}$$

10.3) Polar Coordinates

$$x = r \cos \theta \quad r^2 = x^2 + y^2 \quad \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{r'(\theta) \sin \theta + r(\theta) \cos \theta}{r'(\theta) \cos \theta - r(\theta) \sin \theta}$$

10.4) Areas and Lengths in Polar Coordinates

$$\text{Area of a sector of a circle: } A = \frac{1}{2} r^2 \theta \quad \text{Area of polar region: } A = \frac{1}{2} \int_a^b r^2 d\theta \quad L = \int_a^b \sqrt{r^2 + (\frac{dr}{d\theta})^2} d\theta$$

12.1) 3D - coordinate systems

$$\text{distance formula in 3D: } |P_1 P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

$$\text{eq of sphere: } (x-h)^2 + (y-k)^2 + (z-l)^2 = r^2 \text{ with center } (h, k, l) \text{ and radius } r$$

12.2) Vectors

triangle law  parallelogram law  length: $|a| = \sqrt{a_1^2 + a_2^2} = \sqrt{a_1^2 + a_2^2 + a_3^2}$ vector properties:
 1. $a+b = b+a$ 2. $a+(b+c) = (a+b)+c$ 3. $a+0 = a$ 4. $a+(-a) = 0$ 5. $c(a+b) = ca+cb$ 6. $(c+d)a = ca+da$ 7. $(cd)a = c(da)$ 8. $1a = a$ unit vector: $\hat{u} = \frac{a}{|a|}$

12.3) Dot Product $a \cdot b = a_1 b_1 + a_2 b_2 + a_3 b_3 = |a||b| \cos \theta$

properties of dot product:
 1. $\hat{a} \cdot \hat{a} = |\hat{a}|^2$ 2. $\hat{a} \cdot \hat{b} = \hat{b} \cdot \hat{a}$ 3. $\hat{a} \cdot (\hat{b} + \hat{c}) = \hat{a} \cdot \hat{b} + \hat{a} \cdot \hat{c}$ 4. $(c\hat{a}) \cdot \hat{b} = c(\hat{a} \cdot \hat{b}) = \hat{a} \cdot (c\hat{b})$ 5. $\hat{0} \cdot \hat{a} = 0$
 $\cos \theta = \frac{a \cdot b}{|a||b|}$ orthogonal: $a \cdot b = 0$ same direction: $a \cdot b = |a||b|$ direction angle (w/line): $\cos \alpha = \frac{a_1}{|a|}, \cos \beta = \frac{a_2}{|a|}, \cos \gamma = \frac{a_3}{|a|}$
 $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$
 $\frac{a}{|a|} = \langle \cos \alpha, \cos \beta, \cos \gamma \rangle$

scalar projection b on a: $\text{comp}_a b = \frac{a \cdot b}{|a|}$ vector proj: $\text{proj}_a b = \frac{a \cdot b}{|a|^2} a$

12.4) cross product: $\hat{a} \times \hat{b} = \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{bmatrix}$ the vector $\hat{a} \times \hat{b}$ is orthogonal to both $\hat{a}$ and $\hat{b}$
 $(\hat{a} \times \hat{b}) \cdot \hat{a} = 0$ two vectors are parallel iff $\hat{a} \times \hat{b} = 0$
 $|\hat{a} \times \hat{b}| = \text{area of parallelogram of } \hat{a} \text{ and } \hat{b}$

properties of cross product:
 1. $\hat{a} \times \hat{b} = -\hat{b} \times \hat{a}$ 2. $(c\hat{a}) \times \hat{b} = c(\hat{a} \times \hat{b}) = \hat{a} \times (c\hat{b})$ 3. $\hat{a} \times (\hat{b} + \hat{c}) = \hat{a} \times \hat{b} + \hat{a} \times \hat{c}$ 4. $(\hat{a} + \hat{b}) \times \hat{c} = \hat{a} \times \hat{c} + \hat{b} \times \hat{c}$ 5. $\hat{a} \cdot (\hat{b} \times \hat{c}) = (\hat{a} \times \hat{b}) \cdot \hat{c}$ 6. $\hat{a} \times (\hat{b} \times \hat{c}) = (\hat{a} \cdot \hat{c})\hat{b} - (\hat{a} \cdot \hat{b})\hat{c}$
 triple product: $\hat{a} \cdot (\hat{b} \times \hat{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$
 $i, j, k, \theta = \pi/2$
 $i \times j = k, j \times k = i, k \times i = j$
 $j \times i = -k, k \times j = -i, i \times k = -j$
 volume of parallelepiped: $V = |\hat{a} \cdot (\hat{b} \times \hat{c})|$

12.5) Eqns of Lines and Planes

vector eq: $\vec{r} = \vec{r}_0 + t\vec{v}$ where $\vec{r} = \langle x, y, z \rangle, \vec{r}_0 = \langle x_0, y_0, z_0 \rangle$ vector eq of a plane: $\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$ or $\vec{n} \cdot \vec{r} = \vec{n} \cdot \vec{r}_0$ where $\vec{r} = \langle x, y, z \rangle, \vec{r}_0 = \langle x_0, y_0, z_0 \rangle$
 parametric eq: $x = x_0 + at, y = y_0 + bt, z = z_0 + ct$ scalar eq of a plane: $a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$
 $\hookrightarrow$ line through point (x_0, y_0, z_0) and parallel to direction vector $\langle a, b, c \rangle$ linear eq: $ax + by + cz + d = 0$
 $\hookrightarrow$ 2 vectors are equal iff their components are equal parallel planes: normal vectors are parallel ($\vec{n}_1 \parallel \vec{n}_2$)
 symmetric eqs: $\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$ angle b/w 2 planes: determined by acute angle b/w normals
 line segment from r_0 to r_1 $r(t) = (1-t)r_0 + tr_1, 0 \leq t \leq 1$
 $p = |\text{comp}_{\vec{n}} \vec{b}| = \frac{|\vec{n} \cdot \vec{b}|}{|\vec{n}|} = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$

